

LINEAR PROGRAMMING AND DUALITY

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ABSTRACT. This includes a simple geometrically motivated proof of Farkas' Lemma, and of the LP duality theorem as a consequence, without using the simplex algorithm. See Theorems 4.1 and 3.3. There are no new ideas, but the argument is shorter and clearer than what I have seen elsewhere.

The main source is [BT97, Chapter 4].

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1. Outline of Main Theorem and Proof

This section provides enough information to motivate and construct the proof

See Figure 2. Consider the primal LP problem in standard form:

$$\begin{aligned} & \text{minimise} && cx \\ (1) \quad & \text{subject to} && Ax \geq b, \end{aligned}$$

where $c \in (\mathbb{R}^n)^*$, $x \in \mathbb{R}^n$, A is an $m \times n$ matrix, and $b \in \mathbb{R}^m$.

The dual problem is

$$\begin{aligned} & \text{maximise} && yb \\ (2) \quad & \text{subject to} && yA = c, \ y \geq 0, \end{aligned}$$

where $y \in (\mathbb{R}^m)^*$.

Theorem 1.1.

- (1) For any x and y that are primal and dual feasible respectively, $yb \leq cx$.
- (2) If there is a primal solution x^* then there is a dual solution y^* which is a positive linear combination of the inward normals for the active constraints at x^* . That is, y^* is a positive linear combination of those rows A_i of A such that $A_i x^* = b_i$.
- (3) If x^* and y^* are primal and dual feasible respectively, then $y^* b = c x^*$.

Proof.

- (1) If x and y are primal and dual feasible respectively then $yb \leq yAx = cx$.

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- (2) x^* is a solution of the new primal problem obtained by removing those constraints inactive at x^* . This is clear from Figure 5, which also motivates the proof.

Let $I \subset \{1, \dots, m\}$ denote the active constraints at x^* .

Then the polyhedral cone

$$\mathcal{C} = \{x : A_i x \geq b_i \ \forall i \in I\}$$

is the feasible region for the new primal problem.

Without loss of generality $x^* = 0$. Hence

$$b_i = 0 \text{ for } i \in I, \quad \mathcal{C} = \{x : A_i x \geq 0 \ \forall i \in I\}, \quad cx \geq 0 \text{ for } x \in \mathcal{C}.$$

By definition, c is in the dual cone $\mathcal{C}^*$. But by Farkas' Lemma

$$\mathcal{C}^* = \sum_i \{y_i A_i : i \in I, y_i \geq 0\}$$

We just need “ $\subset$ ” in the previous line, and this is proved using a simple separating hyperplane argument applied to the cone on the right side of the equality. See the proof of Farkas' Lemma. Or see the “gravitational” explanation in Remark 3.1.

By setting $y_i = 0$ in the previous equality if $i \notin I$, it follows for some y^* that

$$c = y^* A, \quad y^* \geq 0, \quad y_i^* = 0 \text{ if } i \notin I.$$

Hence

$$cx^* = y^* A x^* = y^* b,$$

For the last inequality use $A_i x^* = b_i$ if $i \in I$, and $y_i^* = 0$ if $i \notin I$.

- (3) It follows that

$$cx^* = y^* b$$

for *any* solutions x^* and y^* of the primal and dual problems.

□

2. Preliminaries

This section is just basic notation.

Points $x \in \mathbb{R}^n$ are represented by *column* vectors $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$.

For typographic reasons we write $x = (x_1, \dots, x_n)$. Note “ $(,)$ ” instead of “[,]”. For multiplicative purposes we regard x as a $n \times 1$ matrix.

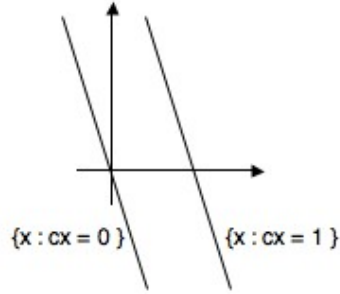
Linear functionals $c \in (\mathbb{R}^n)^*$ are represented by row vectors $c = [c_1 \dots c_n]$. The action on x is given by matrix multiplication

$$cx = \sum_i c_i x_i.$$

Geometrically, if c is non zero then c is represented by a family of parallel hyperplanes, one for each level set of c . See Figure 1.

Sometimes we think of c as a “cost” vector and cx as the cost of the “commodity bundle” x .

Order Relation. By $x \geq 0$ we mean $x_i \geq 0$ for all i . Similarly for $c \geq 0$, etc.


 FIGURE 1. Two level sets corresponding to a linear functional c .

Feasible region. Suppose A is a given $m \times n$ matrix. The rows are denoted A_i . Consider the polyhedral region R , possibly unbounded, given by the constraints

$$R = \{x : Ax \geq b\} = \{x : A_i x \geq b_i \ \forall i = 1, \dots, m\},$$

where $b \in \mathbb{R}^m$ (see Figure 2). We say R is the *feasible* region and elements $x \in R$ are *feasible*. We can also consider constraints of the form $A_i x \leq b_i$ or $A_i x = b_i$

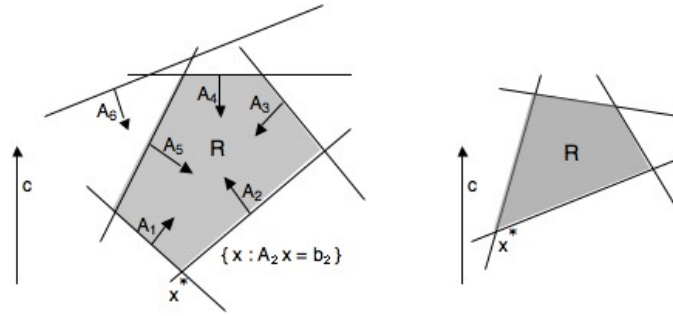


FIGURE 2. The feasible region R consists of those x satisfying $A_i x \geq b_i$ for all i . The normal vectors A_i point “into” R , even for A_6 . The constraints $A_1 x \geq b_1$ and $A_2 x \geq b_2$ are active at x^* , and the other constraints are inactive.

The constraint $A_i \geq b_i$ is *active* at x if $A_i x = b_i$, and otherwise the constraint is *inactive*.

3. Farkas’ Lemma

This is a basic result.

Remark 3.1 (Motivation). See Figure 3.

Think of c as both an element of $\mathbb{R}^n$ and an element of the dual $(\mathbb{R}^n)^*$, in which case $c \cdot x$ corresponds to $c \cdot x$.

Suppose $\mathcal{C}$ is a *polyhedral cone* in $\mathbb{R}^n$ with vertex x^* and is given by the intersection of half spaces $\{x : A_i x \geq b_i\}$ for $i = 1, \dots, m$. By the statement “ x^* is (a) vertex¹ of $\mathcal{C}$ ” is meant that $A_i x^* = b_i$ for all i , i.e. $Ax^* = b$, i.e. *all constraints are active at x^** .

Suppose $\mathcal{C}$ is a subset of the half space through x^* determined by the normal vector c . Then Farkas’ Lemma says c is a linear combination of the inward pointing normals A_i with *non negative* coefficients.

¹The “*” is conventional for minimisers.

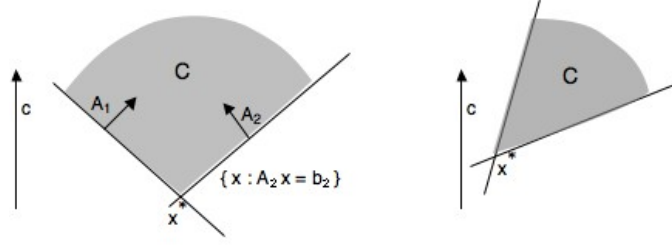


FIGURE 3. In both cases the hypotheses of Farkas' lemma are satisfied. Note that in the second figure $c \notin C$ when the base of c is placed at x^* .

For motivation, suppose gravity acts in the direction $-c$ and consider a small ball B inside the cone C . The ball will be in equilibrium just above the vertex x^* , where the gravitational force is exactly balanced by (inward) normal forces from each of the hyperplanes in contact with B . That is $c = \sum y_i A_i^t$ with $y_i \geq 0$. $\square$

Remark 3.2 (Dual of a Polyhedral Cone). If C is a cone with vertex at the origin, the *dual cone* C^* in $\mathbb{R}^n$ is defined by

$$C^* = \{z \in \mathbb{R}^n : z \cdot x \geq 0 \forall x \in C\}.$$

See Figure 4. Equivalently, working in $(\mathbb{R}^n)^*$, the dual of C is

$$C^* = \{z \in (\mathbb{R}^n)^* : zx \geq 0 \forall x \in C\}.$$

Note $C = C^{**}$.

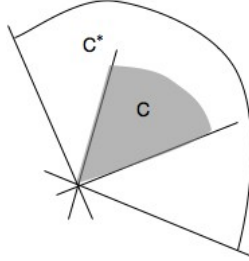


FIGURE 4. A cone C and its dual C^* . Note $C^{**} = C$.

Farkas' Lemma essentially says for a polyhedral cone $C = \{x : Ax \geq 0\}$ that the dual cone is given by

$$C^* = \left\{ \sum_i y_i A_i^t : y_i \geq 0 \right\}.$$

(We take the transpose of the row vectors A_i so as to obtain column vectors in $\mathbb{R}^n$.)

Notationally, it is clearest to prove the equivalent version in $(\mathbb{R}^n)^*$:

$$C^* = \{yA : y \geq 0\}.$$

The inclusion " $\subset$ " was shown in the proof of Farkas' Lemma. The inclusion " $\supset$ " is by the definitions: Consider yA where $y \geq 0$. If $x \in C$, i.e. $Ax \geq 0$, then immediately it follows $yAx \geq 0$. Hence $yA \in C^*$. $\square$

Theorem 3.3 (Farkas' Lemma). *Let C be the cone $\{x : Ax \geq b\} \subset \mathbb{R}^n$ where A is $m \times n$ and the vertex x^* satisfies $Ax^* = b$. Suppose $c \in (\mathbb{R}^n)^*$ and $cx^* \leq cx$ for*

all $x \in \mathcal{C}$. Then there exists $y \in (\mathbb{R}^m)^*$ with

$$c = \sum_i y_i A_i = yA, \quad y \geq 0.$$

(See Figure 3.)

Proof. By translating x^* to the origin we may assume $x^* = 0 \in \mathbb{R}^n$. Hence $b = 0 \in \mathbb{R}^m$, $\mathcal{C} = \{x : Ax \geq 0\}$, and $cx \geq 0$ for $x \in \mathcal{C}$.

We W.T.S. that $c \in \{yA : y \geq 0\} =: \tilde{\mathcal{C}}$.²

Assume $c \notin \tilde{\mathcal{C}}$.

Since $\tilde{\mathcal{C}}$ is a convex set (in fact cone) containing 0 there exists a closed half space containing $\tilde{\mathcal{C}}$ but not containing c . If the half space is given by $z \in \mathbb{R}^n$ then

$$cz < 0, \quad y \geq 0 \implies yAz \geq 0.$$

Since $cz < 0$, $z \notin \mathcal{C}$.

From the implication, $Az \geq 0$, i.e. $z \in \mathcal{C}$.

This contradiction completes the proof. $\square$

Remark 3.4. The vector y^* is not necessarily unique. For example, in $\mathbb{R}^3$, suppose the ball in Remark 3 is supported by 4 or more walls. $\square$

4. Duality in Linear Programming

This is the basic linear programming result.

Consider the linear programming [LP] problem in standard form:

$$(3) \quad \begin{array}{ll} \text{minimise} & cx \\ \text{subject to} & Ax \geq b, \end{array}$$

where $c \in (\mathbb{R}^n)^*$, $x \in \mathbb{R}^n$, A is an $m \times n$ matrix, and $b \in \mathbb{R}^m$.

By straightforward manipulations, linear maximisation problems, linear constraints of the form “ $\leq$ ”, and linear equality constraints, are also included.

Problem 3 is called the *primal problem*. The *dual problem* is then

$$(4) \quad \begin{array}{ll} \text{maximise} & yb \\ \text{subject to} & yA = c, \quad y \geq 0, \end{array}$$

where $y \in (\mathbb{R}^m)^*$.

In the following discussion will see one way in which the dual problem arises naturally.

Theorem 4.1 (LP Duality Theorem).

- (1) Weak Duality: If x is feasible for the primal problem and y for the dual problem then

$$yb \leq cx.$$

- (2) Existence of a Solution of the Dual Problem: If x^* is a solution of the primal problem and y^* is obtained by applying Farkas' Lemma to those rows of A which are active for x^* , then y^* is a solution of the dual problem.

- (3) Strong Duality: If y^* is any solution of the dual problem then

$$y^*b = cx^*.$$

²In fact $\tilde{\mathcal{C}}$ is precisely the dual of $\mathcal{C}$, see Remark 3.2.

Proof. (1) We have

$$\begin{aligned} yb &\leq yAx \quad \text{since } y \geq 0, b \leq Ax \\ &= cx \quad \text{since } yA = c. \end{aligned}$$

(2) By translation assume $x^* = 0$. This will change b but not A .

Suppose A_i is active at 0 for $i \in I$ and inactive for $i \notin I$. Since 0 is feasible,

$$i \in I \implies b_i = 0, \quad i \notin I \implies b_i < 0.$$

Let

$$C = \{x \in \mathbb{R}^n : A_i x \geq 0 \forall i \in I\}.$$

be the cone generated by the constraints active at 0.

We *claim* that if $x \in C$ then ϵx is feasible for the primal problem for sufficiently small $\epsilon > 0$. See Figure 5.

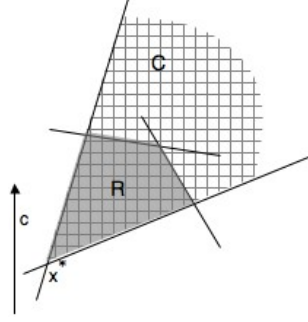


FIGURE 5. C is the cone generated by the active constraints at x^* and R is the feasible region for the LP problem. If x is in the cone given by the active constraints at x^* then ϵx is feasible for sufficiently small $\epsilon > 0$.

If $i \in I$ then $A_i x \geq 0$ and so $A_i(\epsilon x) \geq 0 = b_i$.

If $i \notin I$ then $b_i < 0$ and so $A_i(\epsilon x) \geq b_i$ if $\epsilon > 0$ is sufficiently small.

This establishes the *claim*.

Since $x^* = 0$ minimises the primal problem and ϵx is feasible,

$$c(\epsilon x) \geq cx^* = 0$$

$$\text{and so } cx \geq 0 \text{ if } x \in C.$$

Now apply Farkas' Lemma to deduce

$$c = \sum_{i \in I} y_i^* A_i, \quad \text{where } y_i^* \geq 0.$$

It follows that

$$(5) \quad cx^* = \sum_{i \in I} y_i^* A_i x^* = \sum_{i \in I} y_i^* b_i = y^* b,$$

where we define $y_i^* = 0$ for $i \notin I$. This, together with weak duality, implies y^* maximises the dual problem.

(3) It follows from weak duality and (5) that

$$cx^* = y^*b$$

for *any* solutions x^* and y^* of the primal and dual problems respectively.

□

REFERENCES

- [BT97] Dimitris Bertsimas and John N. Tsitsiklis, *Introduction to Linear Optimization*, Athena Scientific, 1997.